Antonios Kazoulidis

GIS Specialist | Spatial Planning and Development Engineer

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Professional Summary

Spatial Planning and Development Engineer (5-year integrated master's degree from Aristotle University of Thessaloniki, equivalent to MSc) with experience in the preparation, processing, and quality control of geospatial data (raster and vector) using Python and QGIS / ArcGIS Pro. Experienced in the spatial downscaling of grid emissions data to the road network level using datasets from 120+ countries, as well as in documenting deliverables for European research projects. Skilled in utilizing Docker and Linux for reproducible workflows. Native Greek speaker, fluent in English (C2) and basic knowledge of German (A2).

Technical Skills

Geospatial: ArcGIS Pro, QGIS, GDAL/OGR, Leaflet, Coordinate Reference Systems (CRS), Raster and Vector Data Processing, WMS/WFS, GeoJSON, Shapefile, GeoTIFF, KML, Spatial Joins, Clipping, Buffering, AutoCAD

Remote Sensing: Photogrammetry, Agisoft Metashape, ERDAS Imagine, 3D Modeling, Digital Terrain Models (DTM)

Automation & Scripting: Python (geopandas, shapely, pyproj, osmnx, arcpy), Data Quality Control, Claude Code

IT & Databases: SQL, PostGIS, Docker, Linux, Nginx, Git, Microsoft Office

Professional Experience

Urban and Regional Innovation Research Unit (URENIO) Jul 2025 – Jul 2026

Research Associate

- Collected and documented European policies regarding the green transition, digital transformation, and regional innovation for EU research project deliverables.
- Developed thematic maps and geospatial datasets, performing quality and consistency checks prior to final submission.

Hellenic Statistical Authority (ELSTAT) Sep 2021 – Feb 2022

Sector Manager

- Directed a team of four enumerators during the Greek Population and Housing Census of 2021.
- Collected demographic and structural data from 600+ households under strict quality-control protocols.

Recent Projects

Wildfire Scars in Greece Aug 2026

Processed FIRMS VIIRS/MODIS burned area detection data and published as an interactive Leaflet map using GeoJSON layers.

Railway Network by Gauge and Status in Greece Aug 2026

Classified 34 Greek railway lines based on track gauge and operational status using data from Hellenic Train.

Mountainous Designations in Greece Jul 2026

Extracted elevation classes from Digital Terrain Models (DTMs) and compared them against official settlement designations.

Conferences & Presentations

2nd Scientific Conference on Spatial Planning and Development (ESXSA2026) Oct 2026

Presenter

Presented the "GRAVEL" road emissions estimation methodology. Spatial downscaling of road emissions by road segment and vehicle category, processing datasets from 120+ countries using an automated Python workflow.

Education

Aristotle University of Thessaloniki (AUTH) Sep 2020 – Jul 2026

School of Spatial Planning and Development Engineering (Integrated Master, MSc equivalent)

- Diploma Thesis: Achieved a 10/10 grade on the research of spatial downscaling of road transport emission grid data to road network level using GPS Floating Car Data (FCD) and Python.
- Related Coursework: Achieved 10/10 grades in main GIS and spatial analysis modules, such as Geoinformatics Applications, Photogrammetry, Databases, Digital Design, and Urban Planning.